Marvin Wu (Wu Hung Mao)
hongmao.wu@mail.utoronto.ca | LinkedIn | GitHub | (+1)437-970-3837 | Personal website
Research interest: efficient, interpretable video understanding for medical imaging, with weak supervision and selective attention.

Education

Honours Bachelor of Science (HBSc), Computer Science specialist, Mathematics minor. Sep 2021 - Dec 2026
University of Toronto Mississauga (expected graduation with PEY CO-OP - December 2026)
GPA (as of Oct 2026): 3.44/4.0 cumulative | 3.54/4.0 last two years of study (excl. co-op)

Publications

Peer-reviewed publication Sep 2026
Khan, L., **Wu, H.-M.**, Lin, W., Bi, F., Abdelhadi, Y., Jung, J. "Ensemble Deep Learning Approaches for AI-Altered Video Detection." *Journal of Computing, Data, and Exploration (CODEX)*, vol. 2, no. 1, 2026. Preprint: arXiv:2607.06872.

- Presents a multimodal deepfake-detection ensemble (AASIST for audio; EfficientNet-B1, XceptionNet, and MesoNet for video frames) trained on AIGVDBench, FaceForensics++, and ASVspoof 2019 and evaluated on FakeAVCeleb, comparing five fusion strategies.

Work Experience

Teaching Assistant — CSC105H5F (Artificial Intelligence Literacy) Sep 2026 - Dec 2026
University of Toronto *Supervisor: Prof. Josh Jung*

- Hold weekly office hours and provide Piazza support for an introductory AI literacy course of approximately 215 students.
- Prepare and teach 4-hour weekly lab sessions for a 40-student section, grade lab submissions, and invigilate examinations to support course assessment and academic integrity.

Physical Layer (vDU) Developer Intern May 2024 - Sep 2025
Ericsson *Ottawa, Ontario, Canada*

Project: Gphy2.0 (Feb 2025 - Sep 2025)

* Worked alongside senior engineers to move LDPC encoding from an Intel accelerator to NVIDIA GPUs, integrating the GPU branch into the existing codebase.
* Profiled latency of the GPU branch with Nsight Systems and Nsight Compute, comparing reports collected across servers with different GPUs (NVIDIA A10, A100, and Grace Hopper) and adjusting kernels to improve SM occupancy.

Project: Gemini (May 2024 - Feb 2025)

* Digital twin for Radio, RF Channel, and UEs.
* Wrote multi-threaded unit tests in CUDA to test a real-time system under a strict timing constraint ($\sim$ 35 µs).
* Wrote an SQLite script that drew histograms based on data collected from LTTng traces for channel quality analysis.

Research Experience

Early Gastric Neoplasia Grading from Endoscopy Videos Oct 2026 - Present
University of Toronto *Supervisor: Dr. Ameya Pore, Postdoctoral Fellow*

- Joined a project with clinicians building an annotated upper-GI endoscopy video dataset for grading early gastric neoplasia; real-time deployment on the hospital's server is planned as a demonstration.
- Integrated MedSAM2 into the CVAT annotation platform as a GPU-backed interactor and mask tracker, adding mask propagation across endoscopy videos.
- Enabled AMD ROCm GPU acceleration for CVAT's containerized AI models, which only support NVIDIA by default, and ported the MedSAM2 and TransT models from CPU to GPU.
- Built a multi-user tracking service with a job queue, live progress, cancellation, and CVAT-based authentication, processing a 3,790-frame video in about 70 s.
- **Environment:** Docker, ROCm, PyTorch, CVAT, Nuclio; built with AI-assisted development (Claude)

Diffusion Model Research Sep 2026 - Present
University of Toronto *Supervisor: Prof. Samir Hamdi*

- Joined a research project on diffusion models; the research plan and scope are being defined with the professor.

AI-Video Detection Ensemble | *Python, PyTorch, OpenCV, NumPy* Link | Apr 2026
University of Toronto *Supervisor: Prof. Josh Jung*

- Designed a multimodal ensemble that routes faces, video frames, and audio to four different models (EfficientNet, MesoNet, XceptionNet, AASIST) to detect general & deepfake AI-generated video.
- Fine-tuned a pre-trained EfficientNet-B1 model on datasets such as AIGVDBench and FaceForensics++, which use diffusion and face transfer methods to generate deepfake videos.
- Integrated EfficientNet-B1 inference into the ensemble, with MTCNN face cropping and Albumentations preprocessing.
- Built the ensemble's parallel inference pipeline, fusing model scores by mean averaging or stacking (five fusion strategies are compared in the paper) into the final REAL/FAKE prediction; communicated backend results with the frontend through HTTP requests.
- Analyzed the ensemble's ∼70% accuracy on the unseen FakeAVCeleb dataset, against 90%+ in-domain accuracy for individual models such as XceptionNet, and attributed the gap mainly to domain shift.
- Recruited three fellow undergraduates and served as a senior developer on the 5-member team.

Computer Vision & Machine Learning Systems Projects

Multi-Camera Fall Detection (CSC490 Capstone) | *VideoMAE, Gemma 3* Demo | Code | Sep 2026 - Dec 2026

- Built a real-time fall detection system in a 4-person team, where several phone cameras stream to one "room" and vote on each alarm; deployed on Hugging Face Spaces with push alerts to phones.
- Fine-tuned a VideoMAE video transformer on three public fall datasets (EDF, MCFD, UP-Fall) and quantized it to int8 for CPU inference; on held-out recordings replayed as a live stream, it caught 93% of falls a median 1.2 s after onset.
- Required two cameras to agree before raising an alarm, cutting false alarms from 24 to 2 on paired held-out recordings while still catching 94% of falls on those recordings.
- Fine-tuned Gemma 3 12B with LoRA on ∼3,000 clips (H100 GPUs on the Fir cluster) as a second-stage verifier that explains its verdicts and votes alongside a teammate's Qwen2-VL 7B verifier; trained on label-conditioned base-model descriptions, it reached 93.8–96.7% balanced accuracy and removed 5 of 7 false alarms while keeping 132 of 136 falls.
- **Environment:** PyTorch, Hugging Face Transformers, PEFT, Gradio, Modal; built with AI-assisted development (Claude)

Weakly Supervised Polyp Detection in Colonoscopy Video (CSC413) | *DINOv2, Transformers* Oct 2026 - Dec 2026

- Proposed a weakly supervised video transformer that keeps only the top-k most lesion-like patches per frame before temporal attention and trains from video-level labels only, cutting attention cost by ∼1,000× in theory ($k = 8$ of 256 patches).
- Reviewed related work on token selection, multiple instance learning, and video polyp detection, finding few prior methods that combine learned patch selection with weak supervision in endoscopy video.
- Model lead on a 3-person team: co-wrote the NeurIPS-format proposal with baselines and an evaluation plan (video AUROC, frame-level localization, patch hit rate, compute) on public colonoscopy datasets, as a testbed for the gastric neoplasia grading research.

JetBrains LCA Benchmark Contribution | *Python, Docker, OpenAI API, Hugging Face API* Link | Apr 2026

- Contributed to the open-source JetBrains LCA benchmark, expanding the bug localization benchmark's capability from identifying the buggy file to automatically applying a fix.
- Expanded evaluation metrics of the original benchmark by adding token usage count, iteration count, etc. for popular coding agents like Claude Code and Codex.
- Co-authored a technical white paper detailing the multi-agent architecture and evaluation metrics used to benchmark Claude Code and Codex; delivered a group presentation to **40+ attendees**.

Technical Skills

- **Languages:** Python, C, C++, CUDA, Java, SQL, JavaScript
- **ML & Computer Vision:** PyTorch, OpenCV, NumPy, Hugging Face, MTCNN, Albumentations
- **Systems & Profiling:** Linux, Docker, Bazel, Nsight Systems, Nsight Compute, LTTng, DCGM, Valgrind
- **Tools & Infrastructure:** Git/GitHub/GitLab, Gerrit, Jenkins, AWS, HPC cluster (Fir), SQLite, PostgreSQL